# Hyperspectral and Multispectral Image Fusion via Coupled Tensor Wheel Decomposition

Ting Xu, Jin-Liang Xiao, Ya-Ru Fan, Liu Liu, *Senior Member, IEEE*, and Liang-Jian Deng, *Senior Member, IEEE*

***Abstract*—Hyperspectral and multispectral image fusion (HSI-MSI fusion) focuses on recovering a high-resolution hyperspectral image (HSI) by integrating a low-resolution HSI (LR-HSI) with its high-resolution multispectral counterpart. Recent advances in coupled tensor decomposition (CTD)-based approaches have demonstrated strong potential by jointly modeling the intrinsic correlations among the three images within a unified tensor decomposition (TD) framework. However, the TDs employed in existing CTD-based methods provide limited capability in extracting features and capturing intrinsic correlations. In this study, we first examine the feature extraction ability of several representative TD formats. Motivated by the excellent ability of the tensor wheel (TW) decomposition to capture multiway correlations and extract complex features, we develop a new variational HSI-MSI fusion model based on coupled TW decomposition, dubbed coupled tensor wheel decomposition (CTW). We develop a proximal alternating minimization-based algorithm to solve the CTW model. Moreover, we analyze its computational complexity and numerical convergence behavior. Extensive experiments on both simulated and real datasets verify that the proposed CTW approach provides improved fusion results than representative state-of-the-art methods.**

***Index Terms*—Image fusion, proximal alternating minimization, remote sensing, tensor wheel (TW) decomposition, variational model.**

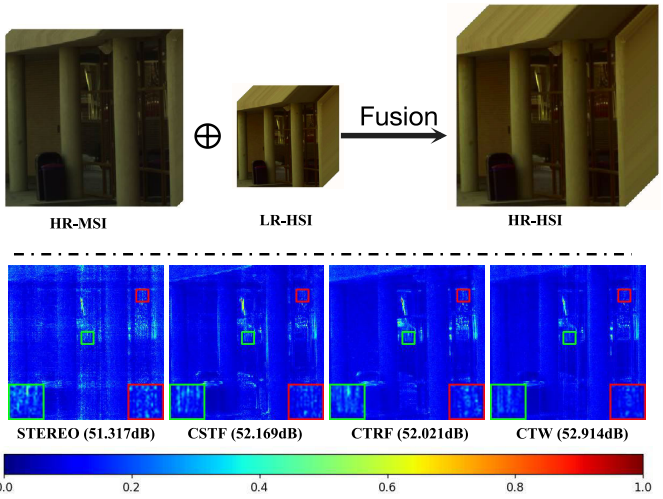



Fig. 1. (Top) Schematic of HSI-MSI fusion task. (Bottom) Error images (30th, 20th, and 10th bands) of the fused *Harvard-imgb9* obtained by different CTD-based methods and the corresponding PSNR values: STEREO [11] (coupled CP), CSTF [12] (coupled Tucker), CTRF [13](coupled TR), and the proposed coupled TW.

## I. Introduction

HYPERSPECTRAL imaging sensors acquire hyperspectral images (HSIs) across a wide range of spectral bands, typically covering a spectral range that extends from the visible region to the shortwave infrared band. The resulting high spectral resolution enables precise material identification and discrimination, allowing subtle differences in material composition to be effectively captured. These advantages have made HSIs widely used in a variety of remote sensing tasks, including vegetation monitoring [1], soil analysis [2], and remote sensing [3], [4], [5], [6], [7], [8], [9], [10]. Nevertheless, constrained by hardware limitations, HSIs typically exhibit low spatial resolution, which constrains the performance of downstream tasks. While enhancing sensor capabilities could improve spatial resolution, this approach is typically expensive and impractical. A cost-efficient alternative is to integrate a low-resolution HSI (LR-HSI) with a high-resolution multispectral image (HR-MSI), which is commonly referred to as hyperspectral and multispectral image fusion (HSI-MSI fusion) (as shown in Fig. 1). The objective of this process is to recover a high-resolution hyperspectral image (HR-HSI) by a pair of LR-HSI and HR-MSI. Existing methods for HSI-MSI fusion can generally be classified into three main categories: 1) deep learning (DL)-based methods; 2) matrix factorization (MF)-based methods; and 3) tensor decomposition (TD)-based methods.

The DL-based approaches, especially the data-driven convolutional neural network-based methods, have shown promising performance [14], [15], [16], [17], [18], [19]. Li et al. [16] introduced a deep image prior-based approach for the detail extraction. Owing to their outstanding feature representation

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Ting Xu is with the School of Mathematical Sciences, Chengdu University of Technology, Chengdu 610059, China (e-mail: tingxu@cdut.edu.cn).

Jin-Liang Xiao is with the School of Mathematics, Southwest Jiaotong University, Chengdu 611756, China (e-mail: jinliang_xiao@163.com).

Liang-Jian Deng are with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: liangjian.deng@uestc.edu.cn).

Ya-Ru Fan is with the School of Mathematics, Southwest Minzu University, Chengdu 610225, China (e-mail: yarufanfan@163.com).

Liu Liu is with the School of Mathematical Sciences and Geomathematics Key Laboratory of Sichuan Province, Chengdu University of Technology, Chengdu 610059, China (e-mail: liuliums@cdut.edu.cn).

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capability, transformer-based methods have garnered significant interest in recent studies. Hu et al. [20] designed a transformer-based framework for HSI-MSI fusion. This idea has been further developed in subsequent works [21], [22], [23], [24], [25], [26], [27]. Fu et al. [22] developed a multiscale transformer-based architecture with spatial–spectral dilated operations for the HSI-MSI fusion. Recently, diffusion-based models have attracted growing interest in HSI-MSI fusion due to their strong generative capability [28], [29], [30], [31], [32]. Cao et al. [29] proposed a supervised diffusion framework equipped with two conditional modulation modules for HSI-MSI fusion. These supervised approaches depend on the "invariance among scale" hypothesis [33], which, however, does not hold reliably in real-world scenarios. To overcome this drawback, unsupervised DL methods are introduced for HSI-MSI fusion [34], [35], [36], [37], [38]. Hong et al. [39] introduced a subpixel-level HSI-MSI fusion framework that adopts a novel decoupled–coupled network to progressively integrate information across pixel- and subpixel-level representations, as well as image- and feature-level domains. However, these methods typically require end-to-end training on large-scale simulated datasets, which is often time-consuming. Extensive research has been conducted to address these challenges [40], [41], [42]. Xiao et al. [42] introduced a guided diffusion strategy that combines zero-shot guidance with neural spatial–spectral decomposition to iteratively generate RGB detail images and transfer these details to HR-HSI, which effectively addresses these two problems. However, the issue of interpretability still poses a challenge in the DL-based methods community. Besides, these methods lack the generality for HSIs from different sources. Once the test data is inconsistent from the training data, their performance is heavily weakened.

The MF-based methods usually decompose the HR-HSI into a spectral basis and coefficient matrix, thereby casting the HSI-MSI fusion problem as a dual-factor estimation task. Some methods [43], [44], [45] argue that the spatial and spectral information are primarily contained in HR-MSI and LR-HSI, respectively. They estimate the coefficient matrix from HR-MSI and obtain the spectral basis from LR-HSI. Some studies [46], [47], [48], [49], [50] argue that the spatial features are also contained in LR-HSI, which can assist in the estimation of the coefficient matrix. Simões et al. [46] estimated the spectral basis by imposing singular value decomposition (SVD) on LR-HSI and formulated a regularized optimization model based on HR-MSI and LR-HSI to obtain the coefficient matrix. Other works [51], [52], [53], [54], [55] argue that the spatial and spectral information are contained both HR-MSI and LR-HSI, and usually update these two matrix factors alternatively. Dian et al. [54] formulated a regularized optimization model based on HR-MSI and LR-HSI to obtain the coefficient matrix and spectral basis. Recently, Xue et al. [55] proposed a HSI-MSI fusion method, called $S^4$-LRR, which first introduced structured sparse low-rank representation (SSLRR), then separately learns the spatial SSLRR from HR-MSI and the spectral SSLRR from LR-HSI. Then, the acquired spatial–spectral SSLRR is integrated into the HSI-MSI fusion framework to regulate the spatial–spectral prior of HR-HSI. The MF-based methods can achieve good fusion accuracy. However, the unfolding operation inevitably breaks the intrinsic 3-D structure of the HR-HSI, which in turn limits their fusion performance.

To overcome the inherent limitation (i.e., the disruption of the 3-D structure of HR-HSI) of the MF-based methods, TD-based techniques have emerged as a promising alternative in preserving the original 3-D structure of HR-HSI. These techniques can be categorized into two families: uncoupled TD (UTD)-based methods and coupled TD (CTD)-based methods. UTD-based methods, e.g., [56], [57], [58], [59], [60], [61], formulate the fusion model in tensor format and utilize the prior information of HR-HSI through TD techniques. For example, Dian et al. [56] designed a low tensor-train rank regularization term to characterize the nonlocal self-similarity (NSS) prior of HR-HSI, called LTTR. Recently, Dian et al. [57] developed a generalized tensor nuclear norm-based HSI-MSI fusion approach, named GTNN, wherein they design a GTNN regularization term to capture the NSS prior of HR-HSI. However, these methods cannot fully characterize the intrinsic structures in HR-MSI and LR-HSI, as they do not explicitly model them within a unified TD framework, which may consequently limit the fusion performance. By contrast, CTD-based methods explicitly represent HR-HSI, HR-MSI, and LR-HSI within a unified TD framework, thereby leveraging structural priors from all three modalities. For example, Kanatsoulis et al. [11] adopted the canonical polyadic (CP) decomposition for HR-HSI, guided by the degradation model, expressed HR-MSI and LR-HSI in CP form. This coupled CP formulation (called STEREO) can capture the low-rank multilinear structure across the three images. Li et al. [12] utilized the Tucker decomposition to model HR-HSI and similarly derived Tucker formats for HR-MSI and LR-HSI from the degradation model, resulting in a coupled Tucker decomposition framework (called CSTF) that exploits multilinear low-rank structure. He et al. [13] follow the same paradigm with the tensor ring (TR) decomposition, deriving a coupled TR method (named CTRF) that encodes multiway low-rank priors. In summary, the choice of TD format is crucial for CTD-based fusion methods. However, the TDs adopted in existing CTD-based approaches offer limited feature extraction capacity and insufficient ability to characterize intrinsic correlations within the images, which consequently leads to suboptimal fusion accuracy.

Recently, Wu et al. [62] proposed a new TD format, termed the tensor wheel (TW) decomposition. Fig. 2 provides a visual illustration of the TW representation for a 3rd-order tensor. Compared with the CP, the Tucker, and the TR decompositions, the TW format offers a more accurate characterization of intrinsic correlations within tensors. In this article, we further examine its feature extraction ability and compare it with the CP, Tucker, and TR formats, as detailed in Section III-B. The results show that the TW decomposition possesses stronger feature extraction capability. Motivated by these findings, we develop a coupled TW decomposition method for HSI-MSI fusion, termed coupled tensor wheel decomposition (CTW). Using the TW representation, the fusion task is reformulated as the estimation of the associated TW factors. Specifically,

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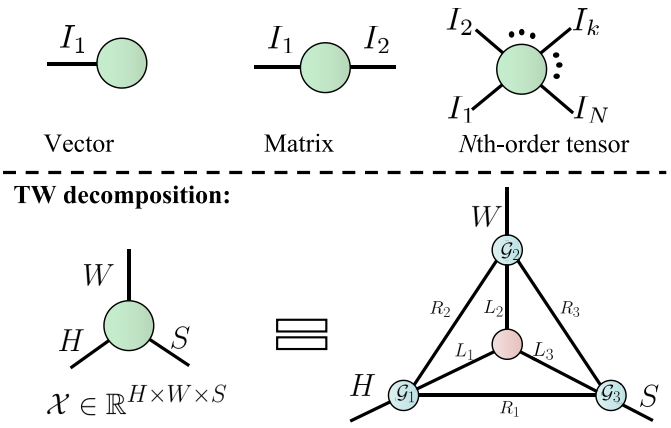



Fig. 2. (Top) Graphical representations of vector, matrix, and $N$th-order tensor. (Bottom) Graphical representation of the TW decomposition for a 3rd-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$.

we develop a degradation model from the HR-HSI to both the HR-MSI and the LR-HSI, as depicted in Fig. 3. In this model, the HR-HSI is degraded to the LR-HSI by downsampling the two 4th-order spatial factors, and to the HR-MSI by downsampling the 4th-order spectral factor. Based on this degradation model, we construct a CTW variational optimization framework and devise a solving algorithm using proximal alternating minimization (PAM) to estimate the TW factors. As shown in Fig. 1, the proposed CTW method achieves superior performance compared with other CTD-based methods. This article makes the following contributions.

1) We observe that the TW decomposition exhibits advantages over the CP, the Tucker, and the TR formats in both feature extraction and the characterization of intrinsic correlations within tensors. Building on these observations, we introduce the TW decomposition into the HSI-MSI fusion task, construct a degradation model based on the TW format, and accordingly formulate a CTW fusion model.
2) We design a PAM-based algorithm to solve the proposed CTW model, and provide a comprehensive analysis of its computational complexity and numerical convergence behavior.
3) Experimental results on three public datasets and one real dataset indicate the superiority of the proposed CTW method over existing HSI-MSI fusion approaches.

The rest of this article is structured as follows. Section II presents the preliminaries. Section III introduces the proposed CTW approach. Section IV reports the numerical experiments. Section V presents the discussion. Finally, Section VI concludes this article.

## II. Preliminaries

In this section, we present the notation and definition related to this work in Section II-A, followed by an overview of the TW decomposition and its key properties in Section II-B.

### A. Work-Related Notation and Definition

The notation adopted in this work is as follows. Bold lowercase symbols (e.g., $\mathbf{x}$) are used for vectors, bold uppercase symbols (e.g., $\mathbf{X}$) are used for matrices, and calligraphic symbols (e.g.,$\mathcal{X}$) are used for tensors. Fig. 2 shows the graphical representations of vector, matrix, and $N$th-order tensor for ease of comparison. For an $M$th-order tensor $\mathcal{Y} \in \mathbb{R}^{J_1\times J_2\times\cdots\times J_M}$, the Frobenius norm is denoted as $\|\mathcal{Y}\|_F = \left(\sum_{j_1 j_2\cdots j_M} \mathcal{Y}(j_1, j_2, \cdots j_M)^2\right)^{1/2}$, where $\mathcal{Y}(j_1, j_2, \ldots, j_M)$ denotes the entry indexed by $(j_1, j_2, \ldots, j_M)$. The notation $(\cdot)^T$ represents matrix transposition, and $\mathbf{I}$ denotes the identity matrix. The definitions pertinent to this study are given as follows.

*Definition 1 (Tensor Permutation [62]):* Let $\mathcal{A} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_M}$ be an $M$th-order tensor and $\mathbf{n} = (n_1, n_2, \ldots, n_M)$ be a permutation of $(1, 2, \ldots, M)$. The $\mathbf{n}$-based permutation of $\mathcal{A}$, denoted by $\mathcal{A}^{\mathbf{n}\rightarrow} \in \mathbb{R}^{I_{n_1}\times I_{n_2}\times\cdots\times I_{n_M}}$, whose entries satisfy

$$\vec{\mathcal{A}}^{\mathbf{n}}\left(i_{n_1}, i_{n_2}, \ldots, i_{n_M}\right) = \mathcal{A}\left(i_1, i_2, \ldots, i_M\right).$$

*Definition 2 (Generalized Tensor k-Unfolding [62]):* Let $\mathbf{n} = (n_1, \ldots, n_M)$ be a permutation of $(1, \ldots, M)$, and let $\mathcal{A} \in \mathbb{R}^{I_1\times\cdots\times I_M}$ be an $M$th-order tensor; the generalized $k$-unfolding $(0 \le k \le M, k \in \mathbb{Z})$ of $\mathcal{A}$ is defined as $\mathbf{A}_{[\mathbf{n};k]} = Unfold_{[\mathbf{n};k]}(\mathcal{A}) \in \mathbb{R}^{\sum_{i=1}^{k} I_{n_i}\times\sum_{j=k+1}^{M} I_{n_j}}$, which satisfies

$$\mathbf{A}_{[\mathbf{n};k]}\left(\overline{i_{n_1} i_{n_2}\cdots i_{n_k}}, \overline{i_{n_{k+1}}\cdots i_{n_M}}\right) = \vec{\mathcal{A}}^{\mathbf{n}}\left(i_{n_1}, i_{n_2}, \ldots, i_{n_M}\right)$$

where $\overline{i_{n_1} i_{n_2}\cdots i_{n_k}}$ and $\overline{i_{n_{k+1}}\cdots i_{n_M}}$ are given by $1 + \sum_{i=1}^{k}(i_{n_i} - 1)\prod_{j=1}^{i-1} I_{n_j}$ and $1 + \sum_{i=k+1}^{M}(i_{n_i} - 1)\prod_{j=k+1}^{i-1} I_{n_j}$, respectively. For the boundary cases $k = 0$ and $k = M$, the resulting vectors, $\mathbf{a}_{[\mathbf{n};0]} \in \mathbb{R}^{1\times\prod_{j=1}^{M} I_{n_j}}$ and $\mathbf{a}_{[\mathbf{n};M]} \in \mathbb{R}^{\prod_{i=1}^{M} I_{n_i}\times 1}$, correspond to two vectorization forms of $\mathcal{A}$. The inverse operator restores the tensor via $\mathcal{A} = Fold_{[\mathbf{n};k]}(\mathbf{A}_{[\mathbf{n};k]})$ or $\mathcal{A} = Fold_{[\mathbf{n};k]}(\mathbf{a}_{[\mathbf{n};k]})$ for $k = 0, M$.

It should be noted that $\mathbf{A}_{[(k,1,2,\ldots,k-1,k+1,\ldots,M);1]}$ and $\mathbf{A}_{[(k,k+1,\ldots,M,1,2,\ldots,k-1);1]}$ are indeed the classical mode-$k$ unfolding $\mathbf{A}_{(k)} \in \mathbb{R}^{I_k\times I_1\cdots I_{(k-1)} I_{(k+1)}\times I_N}$ [63] and mode-$k$ unfolding $\mathbf{A}_{[k]} \in \mathbb{R}^{I_k\times I_{(k+1)}\times I_N I_1\cdots I_{(k-1)}}$ [64], respectively. Their corresponding folding operators are denoted as $\text{Fold}_{(k)}(\cdot)$ and $\text{Fold}_{[k]}(\cdot)$, respectively.

*Definition 3 (Generalized Tensor k-Contraction [62]):* Given an $M$-order tensor $\mathcal{A} \in \mathbb{R}^{I_1\times\cdots\times I_M}$ and an $N$-order tensor $\mathcal{B} \in \mathbb{R}^{J_1\times\cdots\times J_N}$ that share $k$ common modes $(1 \le k \le \min(M, N), k \in \mathbb{Z})$, let $\mathbf{m} = (m_1, m_2, \ldots, m_M)$ and $\mathbf{n} = (n_1, n_2, \ldots, n_N)$ be permutations of $(1, \ldots, M)$ and $(1, \ldots, N)$, respectively, such that $I_{m_i} = J_{n_i}$ for $i = 1, 2, \ldots, k$, and the remaining indices satisfy $m_{k+1} < m_{k+2} < m_M$ and $n_{k+1} < n_{k+2} < n_N$. The generalized $k$-contraction of $\mathcal{A}$ and $\mathcal{B}$ is defined as the $(M + N - 2k)$-order tensor $\mathcal{A}\times_{n_1,n_2,\ldots,n_k}^{m_1,m_2,\ldots,m_k}\mathcal{B} \in \mathbb{R}^{I_{m_{k+1}}\times\cdots\times I_{m_M}\times I_{n_{k+1}}\times\cdots\times I_{n_N}}$, which satisfies

$$\begin{aligned}&\mathcal{A}\times_{n_1,n_2,\ldots,n_k}^{m_1,m_2,\ldots,m_k}\mathcal{B}\\ &= Fold_{[(1,2,\ldots,M+N-2k);M-k]}\left(\mathbf{A}_{[\mathbf{m};k]}^T\mathbf{B}_{[\mathbf{n};k]}\right)\end{aligned}$$

where $\mathbf{A}_{[\mathbf{m};k]}$ and $\mathbf{B}_{[\mathbf{n};k]}$ are the generalized $\mathbf{m}$-based and $\mathbf{n}$-based k-unfolding of tensor $\mathcal{A}$ and $\mathcal{B}$, respectively.

### B. TW Decomposition

The TW decomposition represents $M$th-order tensor $\mathcal{Y} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_M}$ by $M$ 4th-order factors $\mathcal{G}_k \in \mathbb{R}^{R_k\times I_k\times L_k\times R_{k+1}}$,

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$k = 1, 2, \ldots, M$, and an $M$th-order core factor $\mathcal{C} \in \mathbb{R}^{L_1 \times L_2 \times \cdots \times L_M}$, where $R_{M+1}$ is equal to $R_1$. Mathematically, the TW decomposition specifies the element-wise relationship as follows:

$$\mathcal{Y}(i_1, i_2, \ldots, i_M) = \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \cdots \sum_{r_M=1}^{R_M} \sum_{l_1=1}^{L_1} \cdots \sum_{l_M=1}^{L_M} \{\mathcal{G}_1(r_1, i_1, l_1, r_2) \cdots \mathcal{G}_M(r_M, i_M, l_M, r_1)\, \mathcal{C}(l_1, l_2, \ldots, l_M)\}. \quad (1)$$

The tensor form of the TW model (1) is expressed by $\mathcal{X} = \mathcal{G}_1 \times_1^4 \mathcal{G}_2 \times_1^6 \cdots \times_1^{2k} \mathcal{G}_k \times_1^{2(k+1)} \cdots \times_{1,4}^{2M,1} \mathcal{G}_M \times_{1,2,\ldots,M}^{2,4,\ldots,2M} \mathcal{C}$. The TW-ranks is defined by $\mathbf{r} = (R_1, \ldots, R_M, L_1, \ldots, L_M) \in \mathbb{R}^{2M}$. In this article, we adopt the notation $\mathcal{X} = \mathrm{TW}[\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_M; \mathcal{C}]$ to represent TW decomposition. Fig. 2 shows the diagram of the TW decomposition for a third-order tensor. Next, we introduce one important property of TW decomposition.

*Theorem 1:* (Tensor Subwheel Equation [62]:) Assume that the TW decomposition of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is $\overrightarrow{\mathcal{X}}^{\mathbf{n}} = \mathrm{TW}[\mathcal{G}_{n_1}, \mathcal{G}_{n_2}, \ldots, \mathcal{G}_{n_N}; \overrightarrow{\mathcal{C}}^{\mathbf{n}}]$, where $\mathbf{n} = (n_1, n_2, \ldots, n_N)$ denotes an arbitrary circular shift of $(1, 2, \ldots, N)$. Define $\mathbf{m} = (N+1, N+2, 1, 2, \ldots, N)$ and $\mathbf{v} = (2, 4, \ldots, 2N, 1, 3, \ldots, 2N-1)$. Under these settings, the following two tensor subwheel identities hold intrinsically:

$$\mathbf{X}_{[n_N]} = \left(\mathbf{G}_{n_N}\right)_{(2)} \left(\mathbf{M}_{n \neq n_N}\right)_{[\mathbf{m};3]}$$

and

$$\overrightarrow{\mathbf{x}}^{\mathbf{n}}_{[1:N;0]} = \overrightarrow{\mathbf{c}}^{\mathbf{n}}_{[1:N;0]} \left(\mathbf{N}_{\neq \mathcal{C}}\right)_{[\mathbf{v};N]}$$

where $\mathcal{M}_{n \neq n_N} \in \mathbb{R}^{R_{n_1} \times I_{n_1} \times \cdots \times I_{n_{N-1}} \times R_{n_N} \times L_{n_N}}$ is an $(N+2)$th-order subwheel tensor formed by contracting all TW factors except $\mathcal{G}_{n_N}$, that is, $\mathcal{M}_{n \neq n_N} = \mathcal{G}_{n_1} \times_1^4 \cdots \times_1^{2k} \mathcal{G}_{n_k} \times_1^{2k+2} \cdots \times_1^{2N-2} \mathcal{G}_{n_{N-1}} \times_{1,2,\ldots,N-1}^{3,5,\ldots,2N-1} \overrightarrow{\mathcal{C}}^{\mathbf{n}}$. Similarly, $\mathcal{N}_{\neq \mathcal{C}} \in \mathbb{R}^{I_{n_1} \times L_{n_1} \times \cdots \times I_{n_N} \times L_{n_N}}$ is an $2N$th-order subwheel tensor obtained by merging all TW factors except the core tensor $\mathcal{C}$, expressed as $\mathcal{N}_{\neq \mathcal{C}} = \mathcal{G}_{n_1} \times_1^4 \cdots \times_1^{2k} \mathcal{G}_{n_k} \times_1^{2k+2} \cdots \times_{1,4}^{2N,1} \mathcal{G}_{n_N}$.

## III. Proposed Coupled TW Method

In this part, we outline the proposed CTW method. Section III-A introduces the formulation of HSI-MSI fusion problem. Section III-B discusses the motivations for applying the TW decomposition to HSI-MSI fusion. Section III-C presents the proposed CTW method, including the model formulation and the corresponding optimization algorithm.

### A. HSI-MSI Fusion Problem Formulation

In this work, the third-order tensors representing the same scene, HR-HSI, LR-HSI, and HR-MSI, as variables indexed by $(h, w, s)$, where $h$, $w$, and $s$ correspond to the height, the width, and the spectrum dimensions, respectively. The target HR-HSI $\mathcal{X} \in \mathbb{R}^{H \times W \times S}$ has spatial dimensions $H$ and $W$, and spectral dimension $S$. The LR-HSI $\mathcal{Y} \in \mathbb{R}^{h \times w \times S}$ is a spatially degraded version of $\mathcal{X}$, meaning $H > h$ and $W > w$. Meanwhile, the HR-MSI $\mathcal{Z} \in \mathbb{R}^{H \times W \times s}$ suffers from spectral degradation compared to $\mathcal{X}$, with $S > s$. The mathematical formulation of their relationships is given by

$$\begin{aligned} \mathcal{Y} &= \mathcal{X} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2 \\ \mathcal{Z} &= \mathcal{X} \times_3 \mathbf{P}_3 \end{aligned} \quad (2)$$

where $\mathbf{P}_1 \in \mathbb{R}^{h \times H}$, $\mathbf{P}_2 \in \mathbb{R}^{w \times W}$, and $\mathbf{P}_3 \in \mathbb{R}^{s \times S}$ are degradation operators acting on the height, the width, and the spectrum modes of the HR-HSI $\mathcal{X}$, respectively.

The objective of HSI-MSI fusion is to reconstruct the target HR-HSI from the HR-MSI and LR-HSI. The key lies in effectively extracting the spatial and spectral information embedded in these two observations and fusing them in a coherent manner. Motivated by the strong capability of the TW decomposition in feature extraction and in capturing intrinsic correlations within tensors (as analyzed in Section III-B), we employ the TW decomposition to jointly model LR-HSI and HR-MSI under a coupled representation. This formulation enables their features to be absorbed by a shared set of TW factors, which are subsequently used to recover HR-HSI in the TW format.

### B. Motivation of TW Decomposition for HSI-MSI Fusion

The feature extraction capability and the characterization capability of intrinsic tensor correlations of different TD formats are helpful for evaluating their potential in spatial–spectral feature representation. The key motivation behind the proposed CTW method is that TW decomposition offers stronger feature extraction capability and a more effective characterization of intrinsic correlations within tensors compared to the CP, the Tucker, and the TR decompositions.

*1) Feature Extraction Capability:* Here, the TW decomposition is compared with the CP, the Tucker, and the TR format in terms of feature extraction capability. We evaluate their feature extraction capability by the CP-ALS [63], the Tucker-ALS [63], the TR-ALS [64], and the TW-ALS algorithm [62] on the Columbia Object Image Libraries (COIL)-100 dataset.[1] This dataset contains 7200 color images of 100 objects, with 72 images captured per object, exhibiting diverse reflectance properties and complex geometric structures. Each image is a third-order tensor of size $128 \times 128 \times 3$. We downsample each image to $32 \times 32 \times 3$ and organize the dataset so that it is then structured as a third-order tensor $\mathcal{V}$ with dimensions $32^2 \times 3 \times 7200$.

When applying the CP-ALS with a predefined CP-rank $R_1$ to the tensor $\mathcal{V}$, the algorithm produces three factor matrices $\mathbf{U}_i$, $i = 1, 2, 3$, along with a diagonal tensor. Among them, the third factor matrix, i.e., spectral factor $\mathbf{U}_3 \in \mathbb{R}^{R_1 \times 7200}$, is used as the latent CP features. When applying the Tucker-ALS with a predefined Tucker rank $[R_1, R_2, R_3]$ to the tensor $\mathcal{V}$, the algorithm yields three factor matrices $\tilde{\mathbf{U}}_i$, $i = 1, 2, 3$, and a core tensor. The third factor matrix, i.e., spectral factor $\tilde{\mathbf{U}}_3 \in \mathbb{R}^{R_3 \times 7200}$, is extracted as the latent Tucker features. For TR-ALS with a predefined TR-rank $[R_1, R_2, R_3]$, the method generates three 3rd-order TR cores $\hat{\mathcal{G}}_i$, $i = 1, 2, 3$. The third core, i.e., spectral factor $\tilde{\mathcal{G}}_3 \in \mathbb{R}^{R_3 \times 7200 \times R_1}$, is used as the latent TR features. For TW-ALS with a predefined TW-ranks $[R_1, R_2, R_3, L_1, L_2, L_3]$, the algorithm produces three 4th-order ring factors $\mathcal{G}_i$, $i = 1, 2, 3$, together with a 3rd-order core tensor. The third ring factor, i.e., spectral factor $\mathcal{G}_3 \in \mathbb{R}^{R_3 \times 7200 \times L_4 \times R_1}$, is adopted as the latent TW features. In this experiment, for

[1] https://www.kaggle.com/jessicali9530/coil100

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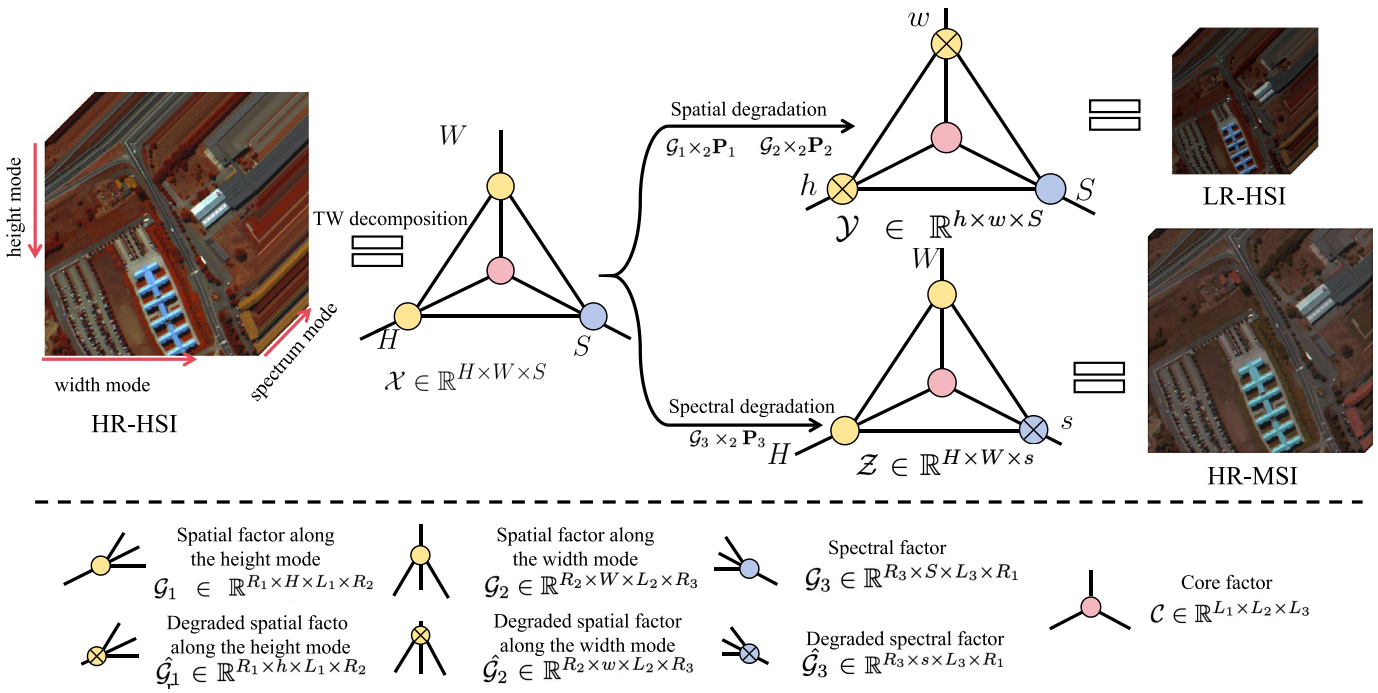



Fig. 3. TW degradation model from the HR-HSI to both LR-HSI and HR-MSI. (Top) From the HR-HSI to LR-HSI spatial degradation. (Bottom) From the HR-HSI to HR-MSI spectral degradation.

the CP-ALS, the Tucker-ALS, the TR-ALS, and the TW-ALS, we set

$$R_1 = R_2 = R_3 = L_1 = L_2 = L_3 = r$$

and vary $r$ from 2 to 3. We set the relative error[2] tolerance and the maximum number of iterations to $10^{-5}$ and 10, respectively. After applying the four ALS algorithms to the tensor $\mathcal{V}$, we regard the CP factor $\mathbf{U}_3$, Tucker factor $\tilde{\mathbf{U}}_3$, TR core $\tilde{\mathcal{G}}_3$, and TW factor $\mathcal{G}_3$ as the extracted latent features. Through permutation and reshaping, the TR core $\tilde{\mathcal{G}}_3$ and the TW factor $\mathcal{G}_3$ are transformed into matrices $\tilde{\mathbf{G}}_3 \in \mathbb{R}^{7200\times r^2}$ and $\mathbf{G}_3 \in \mathbb{R}^{7200\times r^3}$, respectively. We then apply the $k$-nearest neighbors algorithm[3] with $k = 1$ to each of the four matrices, i.e.,$\mathbf{U}_3$ , $\tilde{\mathbf{U}}_3$, $\tilde{\mathbf{G}}_3$, and $\mathbf{G}_3$, to compare the classification accuracy of the four algorithms on the COIL-100 dataset. The results shown in Fig. 4 indicate that the TW-ALS achieves higher classification accuracy than the CP-ALS, the Tucker-ALS, and the TR-ALS, underscoring its superior feature extraction capability.

*2) Characterization of Intrinsic Correlations Within Tensors:* The Tucker topology is encompassed as a special case by the TW model (1) by setting $R_k = 1$ $(k = 1, 2, \ldots, M)$ in the TW-ranks, and the TR topology is obtained when $L_k = 1$ $(k = 1, 2, \ldots, M)$ in the TW-ranks. Moreover, since the CP decomposition is a particular instance of the Tucker format with a diagonal core tensor, it can also be regarded as a degenerate form of the TW model. These relationships naturally explain why the TW decomposition possesses a stronger ability to capture intrinsic tensor correlations than both the Tucker and the TR format.

[2]Relative error = $\|\mathcal{X} - \mathcal{X}_1\|_F / \|\mathcal{X}\|_F$, where $\mathcal{X}$ and $\mathcal{X}_1$ denote the original and reconstructed tensors, respectively.

[3]https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

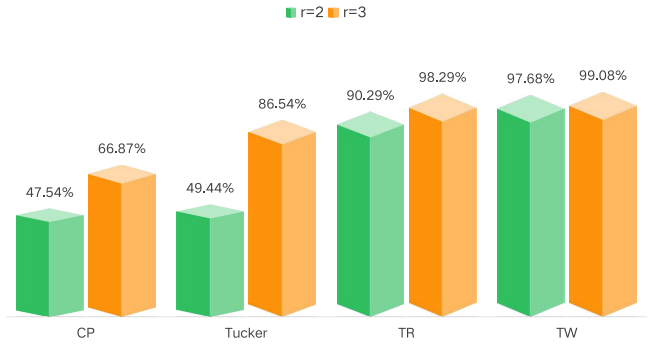



Fig. 4. Classification performance of CP-ALS, Tucker-ALS, TR-ALS, and TW-ALS on the COIL-100 dataset across different tensor ranks.

### C. Proposed CTW Method for HSI-MSI Fusion

Motivated by the merits of the TW decomposition described in Section III-B, we adopt it for the HSI-MSI fusion task. We first represent the target HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ using the TW decomposition as follows:

$$\mathcal{X} = \mathrm{TW}\left[\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3; \mathcal{C}\right] \tag{3}$$

where $\mathcal{G}_1 \in \mathbb{R}^{R_1\times H\times L_1\times R_2}$ and $\mathcal{G}_2 \in \mathbb{R}^{R_2\times W\times L_2\times R_3}$ are the spatial factor along the height and the width modes, respectively; $\mathcal{G}_3 \in \mathbb{R}^{R_3\times S\times L_3\times R_1}$ is the spectral factor; and $\mathcal{C} \in \mathbb{R}^{L_1\times L_2\times L_3}$ is the core tensor connecting the spatial and spectral factors.

Based on the TW decomposition of HR-HSI, i.e., (3), HSI-MSI fusion problem is transferred as the estimation of the $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, and $\mathcal{C}$. Moreover, the degradation model (2) can be reformulated as

$$\begin{aligned}\mathcal{Y} &= \mathrm{TW}\left[\mathcal{G}_1 \times_2 \mathbf{P}_1, \mathcal{G}_2 \times_2 \mathbf{P}_2, \mathcal{G}_3; \mathcal{C}\right] \\ \mathcal{Z} &= \mathrm{TW}\left[\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3 \times_2 \mathbf{P}_3; \mathcal{C}\right]\end{aligned} \tag{4}$$

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where $\mathcal{G}_1 \times_2 \mathbf{P}_1$ and $\mathcal{G}_2 \times_2 \mathbf{P}_2$ are the degraded spatial factor along the height and the width modes, respectively, and $\mathcal{G}_3 \times_2 \mathbf{P}_3$ is the degraded spectral factor. The degradation process is visually illustrated in Fig. 3.

For simplicity, we write $\mathcal{G}_1 \times_2 \mathbf{P}_1$, $\mathcal{G}_2 \times_2 \mathbf{P}_2$, and $\mathcal{G}_3 \times_2 \mathbf{P}_3$ as $\hat{\mathcal{G}}_1 \in \mathbb{R}^{R_1 \times h \times L_1 \times R_2}$, $\hat{\mathcal{G}}_2 \in \mathbb{R}^{R_2 \times w \times L_2 \times R_3}$, and $\hat{\mathcal{G}}_3 \in \mathbb{R}^{R_3 \times s \times L_3 \times R_1}$, respectively. Thus, (4) can be denoted as

$$\begin{aligned} \mathcal{Y} &= \mathrm{TW}\left[\hat{\mathcal{G}}_1, \hat{\mathcal{G}}_2, \mathcal{G}_3; \mathcal{C}\right] \\ \mathcal{Z} &= \mathrm{TW}\left[\mathcal{G}_1, \mathcal{G}_2, \hat{\mathcal{G}}_3; \mathcal{C}\right]. \end{aligned} \tag{5}$$

Based on the degradation model (5), we formulate the following variational CTW optimization model for HSI-MSI fusion:

$$\begin{aligned} \min_{\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3, \mathcal{C}} & \left\|\mathrm{TW}\left[\hat{\mathcal{G}}_1, \hat{\mathcal{G}}_2, \mathcal{G}_3; \mathcal{C}\right] - \mathcal{Y}\right\|_F^2 \\ & + \left\|\mathrm{TW}\left[\mathcal{G}_1, \mathcal{G}_2, \hat{\mathcal{G}}_3; \mathcal{C}\right] - \mathcal{Z}\right\|_F^2. \end{aligned} \tag{6}$$

Simplicity, we denote the objective function in (6) as $f(\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3; \mathcal{C})$, that is, $f(\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3; \mathcal{C}) = \|\mathrm{TW}[\hat{\mathcal{G}}_1, \hat{\mathcal{G}}_2, \mathcal{G}_3; \mathcal{C}] - \mathcal{Y}\|_F^2 + \|\mathrm{TW}[\mathcal{G}_1, \mathcal{G}_2, \hat{\mathcal{G}}_3; \mathcal{C}] -, \mathcal{Z}\|_F^2$. In what follows, we design a solving algorithm for this formulated fusion model.

### D. Algorithm

Directly solving model (6) is challenging because the model is jointly nonconvex, although it is convex with respect to each individual variable. To address this issue, we adopt the PAM scheme, which sequentially updates each variable while holding the others constant. In particular, the tensors $\mathcal{G}_i (i = 1, 2, 3)$ and $\mathcal{C}$ are iteratively updated as follows:

$$\begin{cases} \mathcal{G}_1^{(k+1)} = \min\limits_{\mathcal{G}_1} f\left(\mathcal{G}_1, \mathcal{G}_2^{(k)}, \mathcal{G}_3^{(k)}; \mathcal{C}^{(k)}\right) + \rho \left\|\mathcal{G}_1 - \mathcal{G}_1^{(k)}\right\|_F^2 \\ \mathcal{G}_2^{(k+1)} = \min\limits_{\mathcal{G}_2} f\left(\mathcal{G}_1^{(k+1)}, \mathcal{G}_2, \mathcal{G}_3^{(k)}; \mathcal{C}^{(k)}\right) + \rho \left\|\mathcal{G}_2 - \mathcal{G}_2^{(k)}\right\|_F^2 \\ \mathcal{G}_3^{(k+1)} = \min\limits_{\mathcal{G}_3} f\left(\mathcal{G}_1^{(k+1)}, \mathcal{G}_2^{(k+1)}, \mathcal{G}_3; \mathcal{C}^{(k)}\right) + \rho \left\|\mathcal{G}_3 - \mathcal{G}_3^{(k)}\right\|_F^2 \\ \mathcal{C}^{(k+1)} = \min\limits_{\mathcal{C}} f\left(\mathcal{G}_1^{(k+1)}, \mathcal{G}_2^{(k+1)}, \mathcal{G}_3^{(k+1)}; \mathcal{C}\right) + \rho \left\|\mathcal{C} - \mathcal{C}^{(k)}\right\|_F^2 \end{cases} \tag{7}$$

where $\rho$ is a positive hyperparameter and $(\cdot)^{(k)}$ denotes the estimated blocks of variables in the $k$th iteration. Next, we present the optimizations for $\mathcal{G}_i (i = 1, 2, 3)$ and $\mathcal{C}$ in detail.

*1) $\mathcal{G}_1$ Subproblem:* With $\mathcal{G}_2$, $\mathcal{G}_3$, and $\mathcal{C}$ fixed, $\mathcal{G}_1$ subproblem in (7) can be written as follows:

$$\begin{aligned} \min_{\mathcal{G}_1} & \left\|\mathrm{TW}\left[\hat{\mathcal{G}}_1, \hat{\mathcal{G}}_2^{(k)}, \mathcal{G}_3^{(k)}; \mathcal{C}^{(k)}\right] - \mathcal{Y}\right\|_F^2 \\ & + \left\|\mathrm{TW}\left[\mathcal{G}_1, \mathcal{G}_2^{(k)}, \hat{\mathcal{G}}_3^{(k)}; \mathcal{C}^{(k)}\right] - \mathcal{Z}\right\|_F^2 + \rho \left\|\mathcal{G}_1 - \mathcal{G}_1^{(k)}\right\|_F^2. \end{aligned} \tag{8}$$

Based on Theorem 1, the $\mathcal{G}_1$ subproblem in (8) can be represented as

$$\begin{aligned} \min_{(\mathbf{G}_1)_{(2)}} & \left\|\left(\hat{\mathbf{G}}_1\right)_{(2)} \left(\tilde{\mathbf{M}}_{\neq 1}\right)_{[\mathbf{m};3]} - \mathbf{Y}_{[1]}\right\|_F^2 \\ & + \left\|(\mathbf{G}_1)_{(2)} (\mathbf{M}_{\neq 1})_{[\mathbf{m};3]} - \mathbf{Z}_{[1]}\right\|_F^2 \\ & + \rho \left\|(\mathbf{G}_1)_{(2)} - \left(\mathbf{G}_1^{(k)}\right)_{(2)}\right\|_F^2 \end{aligned} \tag{9}$$

where $\mathbf{Y}_{[1]} \in \mathbb{R}^{h \times wS}$ and $\mathbf{Z}_{[1]} \in \mathbb{R}^{H \times Ws}$ are the mode-1 unfolding of $\mathcal{Y}$ and $\mathcal{Z}$, respectively. $(\hat{\mathbf{G}}_1)_{(2)} = \mathbf{P}_1 (\mathbf{G}_1)_{(2)} \in \mathbb{R}^{h \times R_1 L_1 R_2}$, $(\mathbf{G}_1)_{(2)} \in \mathbb{R}^{H \times R_1 L_1 R_2}$ and $(\mathbf{G}_1^{(k)})_{(2)} \in \mathbb{R}^{H \times R_1 L_1 R_2}$ are the classic mode-2 unfolding of $\mathcal{G}_1$ and $\mathcal{G}_1^{(k)}$, respectively. $(\tilde{\mathbf{M}}_{\neq 1})_{[\mathbf{m};3]} \in \mathbb{R}^{R_1 L_1 R_2 \times wS}$ and $(\mathbf{M}_{\neq 1})_{[\mathbf{m};3]} \in \mathbb{R}^{R_1 L_1 R_2 \times Ws}$ are the tensor 3-unfolding of $\tilde{\mathcal{M}}_{\neq 1} = \hat{\mathcal{G}}_2^{(k)} \times_1^4 \mathcal{G}_3^{(k)} \times_{1,2}^{3,5} (\overrightarrow{\mathcal{C}^{(k)}})^{\mathbf{n}}$ and $\mathcal{M}_{\neq 1} = \mathcal{G}_2^{(k)} \times_1^4 \hat{\mathcal{G}}_3^{(k)} \times_{1,2}^{3,5} (\overrightarrow{\mathcal{C}^{(k)}})^{\mathbf{n}}$, respectively, wherein $\mathbf{m} = [4, 5, 1, 2, 3]$ and $\mathbf{n} = [2, 3, 1]$.

Problem (9) is quadratic and its unique solution amounts to compute the following problem:

$$\begin{aligned} & \mathbf{P}_1^T \mathbf{P}_1 (\mathbf{G}_1)_{(2)} \left(\tilde{\mathbf{M}}_{\neq 1}\right)_{[\mathbf{m};3]} \left(\tilde{\mathbf{M}}_{\neq 1}\right)_{[\mathbf{m};3]}^T \\ & \quad + (\mathbf{G}_1)_{(2)} (\mathbf{M}_{\neq 1})_{[\mathbf{m};3]} (\mathbf{M}_{\neq 1})_{[\mathbf{m};3]}^T + \rho (\mathbf{G}_1)_{(2)} \\ & = \mathbf{P}_1^T \mathbf{Y}_{[1]} \left(\tilde{\mathbf{M}}_{\neq 1}\right)_{[\mathbf{m};3]}^T + \mathbf{Z}_{[1]} (\mathbf{M}_{\neq 1})_{[\mathbf{m};3]}^T + \rho \left(\mathbf{G}_1^{(k)}\right)_{(2)}. \end{aligned} \tag{10}$$

We use the conjugate gradient (CG) method [65] to solve (10). Then, $\mathcal{G}_1 = \mathrm{Fold}_{(2)}\{(\mathbf{G}_1)_{(2)}\}$.

*2) $\mathcal{G}_2$ Subproblem:* With $\mathcal{G}_1$, $\mathcal{G}_3$ and $\mathcal{C}$ fixed, $\mathcal{G}_2$ subproblem in (7) can be written as follows:

$$\begin{aligned} \min_{\mathcal{G}_2} & \left\|\mathrm{TW}\left[\hat{\mathcal{G}}_1^{(k+1)}, \mathcal{G}_2, \mathcal{G}_3^{(k)}; \mathcal{C}^{(k)}\right] - \mathcal{Y}\right\|_F^2 \\ & + \left\|\mathrm{TW}\left[\mathcal{G}_1^{(k+1)}, \mathcal{G}_2, \hat{\mathcal{G}}_3^{(k)}; \mathcal{C}^{(k)}\right] - \mathcal{Z}\right\|_F^2 + \rho \left\|\mathcal{G}_2 - \mathcal{G}_2^{(k)}\right\|_F^2. \end{aligned} \tag{11}$$

Based on Theorem 1, the $\mathcal{G}_2$ subproblem (11) can be represented as

$$\begin{aligned} \min_{(\mathbf{G}_2)_{(2)}} & \left\|\left(\hat{\mathbf{G}}_2\right)_{(2)} \left(\tilde{\mathbf{M}}_{\neq 2}\right)_{[\mathbf{m};3]} - \mathbf{Y}_{[2]}\right\|_F^2 \\ & + \left\|(\mathbf{G}_2)_{(2)} (\mathbf{M}_{\neq 2})_{[\mathbf{m};3]} - \mathbf{Z}_{[2]}\right\|_F^2 \\ & + \rho \left\|(\mathbf{G}_2)_{(2)} - \left(\mathbf{G}_2^{(k)}\right)_{(2)}\right\|_F^2 \end{aligned} \tag{12}$$

where $\mathbf{Y}_{[2]} \in \mathbb{R}^{w \times Sh}$ and $\mathbf{Z}_{[2]} \in \mathbb{R}^{W \times sH}$ are the mode-2 unfolding of $\mathcal{Y}$ and $\mathcal{Z}$, respectively. $(\hat{\mathbf{G}}_2)_{(2)} = \mathbf{P}_2 (\mathbf{G}_2)_{(2)} \in \mathbb{R}^{w \times R_2 L_2 R_3}$, $(\mathbf{G}_2)_{(2)} \in \mathbb{R}^{W \times R_2 L_2 R_3}$ and $(\mathbf{G}_2^{(k)})_{(2)} \in \mathbb{R}^{W \times R_2 L_2 R_3}$ are the classic mode-2 unfolding of $\mathcal{G}_2$ and $\mathcal{G}_2^{(k)}$, respectively. $(\tilde{\mathbf{M}}_{\neq 2})_{[\mathbf{m};3]} \in \mathbb{R}^{R_2 L_2 R_3 \times Sh}$ and $(\mathbf{M}_{\neq 2})_{[\mathbf{m};3]} \in \mathbb{R}^{R_2 L_2 R_3 \times sH}$ are the tensor 3-unfolding of $\tilde{\mathcal{M}}_{\neq 2} = \mathcal{G}_3^{(k)} \times_1^4 \hat{\mathcal{G}}_1^{(k+1)} \times_{1,2}^{3,5} (\overrightarrow{\mathcal{C}^{(k)}})^{\mathbf{n}}$ and $\mathcal{M}_{\neq 2} = \hat{\mathcal{G}}_3^{(k)} \times_1^4 \mathcal{G}_1^{(k+1)} \times_{1,2}^{3,5} (\overrightarrow{\mathcal{C}^{(k)}})^{\mathbf{n}}$, respectively, wherein $\mathbf{m} = [4, 5, 1, 2, 3]$ and $\mathbf{n} = [3, 1, 2]$.

Problem (12) is quadratic and its unique solution amounts to compute the following problem:

$$\begin{aligned} & \mathbf{P}_2^T \mathbf{P}_2 (\mathbf{G}_2)_{(2)} \left(\tilde{\mathbf{M}}_{\neq 2}\right)_{[\mathbf{m};3]} \left(\tilde{\mathbf{M}}_{\neq 2}\right)_{[\mathbf{m};3]}^T \\ & \quad + (\mathbf{G}_2)_{(2)} (\mathbf{M}_{\neq 2})_{[\mathbf{m};3]} (\mathbf{M}_{\neq 2})_{[\mathbf{m};3]}^T + \rho (\mathbf{G}_2)_{(2)} \\ & = \mathbf{P}_2^T \mathbf{Y}_{[2]} \left(\tilde{\mathbf{M}}_{\neq 2}\right)_{[\mathbf{m};3]}^T + \mathbf{Z}_{[2]} (\mathbf{M}_{\neq 2})_{[\mathbf{m};3]}^T + \rho \left(\mathbf{G}_2^{(k)}\right)_{(2)}. \end{aligned} \tag{13}$$

We use the CG algorithm to solve (13). Then, $\mathcal{G}_2 = \mathrm{Fold}_{(2)}\{(\mathbf{G}_2)_{(2)}\}$.

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*3) $\mathcal{G}_3$ Subproblem:* With $\mathcal{G}_1$, $\mathcal{G}_2$ and $\mathcal{C}$ fixed, $\mathcal{G}_3$ subproblem in (7) can be written as follows:

$$\min_{\mathcal{G}_3} \left\| \mathrm{TW}\left[\hat{\mathcal{G}}_1^{(k+1)}, \hat{\mathcal{G}}_2^{(k+1)}, \mathcal{G}_3; \mathcal{C}^{(k)}\right] - \mathcal{Y} \right\|_F^3 + \left\| \mathrm{TW}\left[\mathcal{G}_1^{(k+1)}, \mathcal{G}_2^{(k+1)}, \hat{\mathcal{G}}_3; \mathcal{C}^{(k)}\right] - \mathcal{Z} \right\|_F^3 + \rho \left\| \mathcal{G}_3 - \mathcal{G}_3^{(k)} \right\|_F^2. \tag{14}$$

Based on Theorem 1, the $\mathcal{G}_3$ subproblem (14) can be represented as

$$\min_{(\mathbf{G}_3)_{(2)}} \left\| (\mathbf{G}_3)_{(2)} \left(\tilde{\mathbf{M}}_{\neq 3}\right)_{[\mathbf{m};3]} - \mathbf{Y}_{[3]} \right\|_F^2 + \left\| (\hat{\mathbf{G}}_3)_{(2)} (\mathbf{M}_{\neq 3})_{[\mathbf{m};3]} - \mathbf{Z}_{[3]} \right\|_F^2 + \rho \left\| (\mathbf{G}_3)_{(2)} - \left(\mathbf{G}_3^{(k)}\right)_{(2)} \right\|_F^2 \tag{15}$$

where $\mathbf{Y}_{[3]} \in \mathbb{R}^{S \times hw}$ and $\mathbf{Z}_{[3]} \in \mathbb{R}^{s \times HW}$ are the mode-3 unfolding of $\mathcal{Y}$ and $\mathcal{Z}$, respectively. $(\hat{\mathbf{G}}_3)_{(2)} = \mathbf{P}_3 (\mathbf{G}_3)_{(2)} \in \mathbb{R}^{s \times R_3 L_3 R_1}$, and $(\mathbf{G}_3)_{(2)} \in \mathbb{R}^{S \times R_3 L_3 R_1}$ and $(\mathbf{G}_3^{(k)})_{(2)} \in \mathbb{R}^{S \times R_3 L_3 R_1}$ are the mode-2 unfolding of $\mathcal{G}_3$ and $\mathcal{G}_3^{(k)}$, respectively. $(\tilde{\mathbf{M}}_{\neq 3})_{[\mathbf{m};3]} \in \mathbb{R}^{R_3 L_3 R_1 \times hw}$ and $(\mathbf{M}_{\neq 3})_{[\mathbf{m};3]} \in \mathbb{R}^{R_3 L_3 R_1 \times HW}$ are the tensor 3-unfolding of $\tilde{\mathcal{M}}_{\neq 3} = \hat{\mathcal{G}}_1^{(k+1)} \times_1^4 \hat{\mathcal{G}}_2^{(k+1)} \times_{1,2}^{3,5} (\overrightarrow{\mathcal{C}^{(k)}})^{\mathbf{n}}$ and $\mathcal{M}_{\neq 3} = \mathcal{G}_1^{(k+1)} \times_1^4 \mathcal{G}_2^{(k+1)} \times_{1,2}^{3,5} (\overrightarrow{\mathcal{C}^{(k)}})^{\mathbf{n}}$, respectively, wherein $\mathbf{m} = [4, 5, 1, 2, 3]$ and $\mathbf{n} = [1, 2, 3]$.

Problem (15) is quadratic and its unique solution amounts to compute the following problem:

$$\mathbf{P}_3^T \mathbf{P}_3 (\mathbf{G}_3)_{(2)} (\mathbf{M}_{\neq 3})_{[\mathbf{m};3]} (\mathbf{M}_{\neq 3})_{[\mathbf{m};3]}^T + (\mathbf{G}_3)_{(2)} \left(\tilde{\mathbf{M}}_{\neq 3}\right)_{[\mathbf{m};3]} \left(\tilde{\mathbf{M}}_{\neq 3}\right)_{[\mathbf{m};3]}^T + \rho (\mathbf{G}_3)_{(2)} = \mathbf{P}_3^T \mathbf{Z}_{[3]} (\mathbf{M}_{\neq 3})_{[\mathbf{m};3]}^T + \mathbf{Y}_{[3]} \left(\tilde{\mathbf{M}}_{\neq 3}\right)_{[\mathbf{m};3]}^T + \rho \left(\mathbf{G}_3^{(k)}\right)_{(3)}. \tag{16}$$

We use the CG algorithm to solve (16). Then, $\mathcal{G}_3 = \mathrm{Fold}_{(2)}\{(\mathbf{G}_3)_{(2)}\}$.

*4) $\mathcal{C}$ Subproblem:* With $\mathcal{G}_1$, $\mathcal{G}_2$, and $\mathcal{G}_3$ fixed, the $\mathcal{C}$ subproblem in (7) can be written as follows:

$$\min_{\mathcal{C}} \left\| \mathrm{TW}\left[\hat{\mathcal{G}}_1^{(k+1)}, \hat{\mathcal{G}}_2^{(k+1)}, \mathcal{G}_3^{(k+1)}; \mathcal{C}\right] - \mathcal{Z} \right\|_F^2 + \left\| \mathrm{TW}\left[\mathcal{G}_1^{(k+1)}, \mathcal{G}_2^{(k+1)}, \hat{\mathcal{G}}_3^{(k+1)}; \mathcal{C}\right] - \mathcal{Y} \right\|_F^2 + \rho \left\| \mathcal{C} - \mathcal{C}^{(k)} \right\|_F^2. \tag{17}$$

Based on Theorem 1, the $\mathcal{C}$ subproblem (17) can be represented as

$$\min_{\mathcal{C}} \left\| \mathbf{c}_{[1:3;0]} \left(\tilde{\mathbf{M}}_{\neq \mathcal{C}}\right)_{[\mathbf{v};3]} - \mathbf{y}_{[1:3;0]} \right\|_F^2 + \left\| \mathbf{c}_{[1:3;0]} (\mathbf{M}_{\neq \mathcal{C}})_{[\mathbf{v};3]} - \mathbf{z}_{[1:3;0]} \right\|_F^2 + \rho \left\| \mathbf{c}_{[1:3;0]} - \mathbf{c}_{[1:3;0]}^{(k)} \right\|_F^2 \tag{18}$$

where $\mathbf{y}_{[1:3;0]} \in \mathbb{R}^{1 \times hwS}$, $\mathbf{z}_{[1:3;0]} \in \mathbb{R}^{1 \times HWs}$, $\mathbf{c}_{[1:3;0]} \in \mathbb{R}^{1 \times L_1 L_2 L_3}$, and $\mathbf{c}_{[1:3;0]}^{(k)}$ are the vectorizations of $\mathcal{Y}$, $\mathcal{Z}$, $\mathcal{C}$, and $\mathcal{C}^{(k)}$, respectively. $(\tilde{\mathbf{M}}_{\neq \mathcal{C}})_{[\mathbf{v};3]} \in \mathbb{R}^{L_1 L_2 L_3 \times hwS}$ and $(\mathbf{M}_{\neq \mathcal{C}})_{[\mathbf{v};3]} \in \mathbb{R}^{L_1 L_2 L_3 \times HWs}$ are the generalized tensor 3-unfolding of $\tilde{\mathcal{M}}_{\neq \mathcal{C}} = (\hat{\mathcal{G}}_1^{(k+1)} \times_1^4 \hat{\mathcal{G}}_2^{(k+1)} \times_{1,4}^{6,1} \mathcal{G}_3^{(k+1)})$ and $\mathcal{M}_{\neq \mathcal{C}} = (\mathcal{G}_1^{(k+1)} \times_1^4 \mathcal{G}_2^{(k+1)} \times_{1,4}^{6,1} \hat{\mathcal{G}}_3^{(k+1)})$, respectively, wherein $\mathbf{v} = (2, 4, 6, 1, 3, 5)$.

Problem (18) is quadratic and its unique solution is

$$\mathbf{c}_{[1:3;0]} = \left(\mathbf{y}_{[1:3;0]} \left(\tilde{\mathbf{M}}_{\neq \mathcal{C}}\right)_{[\mathbf{v};3]}^T + \mathbf{z}_{[1:3;0]} (\mathbf{M}_{\neq \mathcal{C}})_{[\mathbf{v};3]}^T + \rho \mathbf{c}_{[1:3;0]}^{(k)}\right) / \times \left(\left(\tilde{\mathbf{M}}_{\neq \mathcal{C}}\right)_{[\mathbf{v};3]} \left(\tilde{\mathbf{M}}_{\neq \mathcal{C}}\right)_{[\mathbf{v};3]}^T + (\mathbf{M}_{\neq \mathcal{C}})_{[\mathbf{v};3]} (\mathbf{M}_{\neq \mathcal{C}})_{[\mathbf{v};3]}^T + \rho \mathbf{I}\right). \tag{19}$$

Then, we obtain the core tensor $\mathcal{C}$ by $\mathcal{C} = Fold_{[1:3;0]}(\mathbf{c}_{[1:3;0]})$.

We outline the proposed CTW method for HSI-MSI fusion in Algorithm 1. The algorithm solves the CTW model (6) using the PAM scheme described in (7). Because the optimization problem (6) is jointly nonconvex, a well-designed initialization is essential to avoid undesirable local minima. Considering that the HR-MSI mainly preserves spatial structures, while the LR-HSI retains richer spectral content, we initialize the spatial factors $\mathcal{G}_1$ and $\mathcal{G}_2$ using the HR-MSI and the spectral factor $\mathcal{G}_3$ from the LR-HSI. The factors $\mathcal{G}_1$ and $\mathcal{G}_2$ are initialized using the TW-ALS algorithm [62], providing a stable and accurate starting point for the spatial domains. The tensor $\mathcal{G}_3$ is initialized using the simplex identification split augmented Lagrangian (SISAL) method [66], which efficiently identifies a minimum volume simplex containing the LR-HSI spectral vectors. With both the spectral and spatial bases well-anchored, a standard stochastic initialization is sufficient for the core tensor $\mathcal{C}$. A flowchart of the full CTW fusion framework is provided in Fig. 5.

### *E. Computational Complexity Analysis*

In Algorithm 1, we optimize the three factors $\mathcal{G}_i (i = 1, 2, 3)$ by the CG algorithm and the core tensor $\mathcal{C}$ by the matrix multiplication. In each iteration of CG, the most computationally demanding operation is multiplying the system matrix with a vector. For the optimization with respect to $(\mathbf{G}_1)_{(2)}$, the complexity is $\mathcal{O}(H^2(R_1 L_1 R_2)^2)$, with respect to $(\mathbf{G}_2)_{(2)}$, the complexity is $\mathcal{O}(W^2(R_2 L_2 R_3)^2)$, and with respect to $(\mathbf{G}_3)_{(2)}$, the complexity is $\mathcal{O}(S^2(R_3 L_3 R_1)^2)$. These operations may be carried out efficiently in a matrix representation, reducing the complexity to $\mathcal{O}(H^2(R_1 L_1 R_2) + H(R_1 L_1 R_2)^2)$, $\mathcal{O}(W^2(R_2 L_2 R_3) + W(R_2 L_2 R_3)^2)$, and $\mathcal{O}(S^2(R_3 L_3 R_1) + S(R_3 L_3 R_1)^2)$. The core tensor is updated by (19), and its complexity is $\mathcal{O}((L_1 L_2 L_3)^2(hwS + HWs))$. Thus, the total complexity of each iteration in Algorithm 1 is

$$\mathcal{O}\left(N_{CG1}\left(H^2(R_1 L_1 R_2) + H(R_1 L_1 R_2)^2\right) + N_{CG2}\left(W^2(R_2 L_2 R_3) + W(R_2 L_2 R_3)^2\right) + N_{CG3}\left(S^2(R_3 L_3 R_1) + S(R_3 L_3 R_1)^2\right) + (L_1 L_2 L_3)^2(hwS + HWs)\right) \tag{20}$$

where $N_{CG1}$, $N_{CG2}$, and $N_{CG3}$ denote the number of CG iterations for updating $(\mathbf{G}_1)_{(2)}$, $(\mathbf{G}_2)_{(2)}$, and $(\mathbf{G}_3)_{(2)}$, respectively. In our experiments, we found that 30 iterations yield a good approximation of the solution for (10), (13), and (16). Therefore, we set $N_{CG1} = N_{CG2} = N_{CG3} = 30$.

Here, we also provide the per-iteration time complexity of the coupled Tucker decomposition method (i.e., CSTF) to

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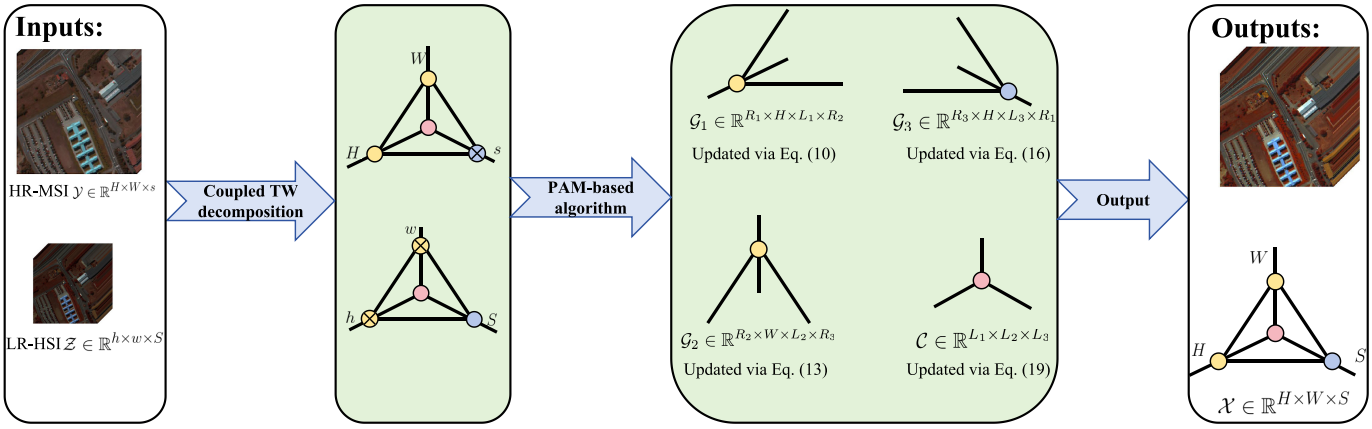



Fig. 5. Overall framework of the proposed CTW method for HSI-MSI fusion. Specifically, the input HR-MSI ($\mathcal{Y}$) and LR-HSI ($\mathcal{Z}$) are first represented in their corresponding TW formats. Subsequently, the underlying TW factors (including $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, and the core tensor $\mathcal{C}$) are iteratively updated using the proposed PAM-based algorithm. Finally, the target HR-HSI ($\mathcal{X}$) is reconstructed from the optimized TW factors.

**Algorithm 1** Proposed CTW Method for HSI-MSI Fusion

**Input:** HR-MSI $\mathcal{Z}$, LR-HSI $\mathcal{Y}$, the predefined TW-ranks $\mathbf{r}$, the positive hyperparameter $\rho$, the number of iterations $K$.
**Output:** HR-HSI $\mathcal{X}$.

Initialize $k = 0$. Set $\mathcal{G}_1^{(0)}$ and $\mathcal{G}_2^{(0)}$ using the TW-ALS algorithm applied to $\mathcal{Z}$, obtain $\mathcal{G}_3^{(0)}$ from $\mathcal{Y}$ using the SISAL algorithm, and initialize $\mathcal{C}^0$ with random values.
**while** $k < K$ **do**
  Update $(\mathbf{G}_1^{(k+1)})_{(2)}$ via (10) with CG, then $\mathcal{G}_1^{(k+1)} = Fold_{(2)}\{(\mathbf{G}_1^{(k+1)})_{(2)}\}$.
  Update $(\mathbf{G}_2^{(k+1)})_{(2)}$ via (13) with CG, then $\mathcal{G}_2^{(k+1)} = Fold_{(2)}\{(\mathbf{G}_2^{(k+1)})_{(2)}\}$.
  Update $(\mathbf{G}_3^{(k+1)})_{(2)}$ via (16) with CG, then $\mathcal{G}_3^{(k+1)} = Fold_{(2)}\{(\mathbf{G}_3^{(k+1)})_{(2)}\}$.
  Update $\mathbf{c}_{[1:3;0]}^{(k+1)}$ via (19), then $\mathcal{C}^{(k+1)} = Fold_{[1:3;0]}\{\mathbf{c}_{[1:3;0]}^{(k+1)}\}$.
  $k \leftarrow k + 1$.
**end while**
Obtain HR-HSI $\mathcal{X}$ via (3).

TABLE I
RESULTS FOR FIVE METRICS ACROSS ALL METHODS ARE PRESENTED FOR THE CAVE, HARVARD, AND CHIKUSEI DATASETS (BOLD: BEST; UNDERLINE: SECOND BEST)

| Dataset | Method | STEREO [11] | CSTF [12] | LTTR [56] | CTRF [13] | GTNN[57] | CTW (Ours) |
|---|---|---|---|---|---|---|---|
| *CAVE-flowers* | PSNR ↑ | 41.209 | 47.277 | 40.375 | 47.085 | 40.621 | **47.599** |
| | DD ↓ | 2.023 | 0.810 | 1.186 | 0.808 | 1.156 | **0.775** |
| | ERGAS ↓ | 4.030 | 1.844 | 3.358 | 1.829 | 3.255 | **1.737** |
| | SSIM ↑ | 0.927 | 0.987 | 0.984 | 0.986 | 0.985 | **0.988** |
| | RMSE ↓ | 3.817 | 1.461 | 2.885 | 1.470 | 2.815 | **1.430** |
| *Harvard-imgb9* | PSNR ↑ | 51.317 | 52.169 | 45.732 | 52.021 | 46.108 | **52.914** |
| | DD ↓ | 0.581 | 0.481 | 0.999 | 0.487 | 0.957 | **0.457** |
| | ERGAS ↓ | 0.828 | 0.804 | 1.309 | 0.785 | 1.216 | **0.686** |
| | SSIM ↑ | 0.995 | 0.992 | 0.991 | **0.996** | 0.992 | **0.996** |
| | RMSE ↓ | 0.851 | 0.668 | 1.696 | 0.704 | 1.659 | **0.655** |
| *Chikusei* | PSNR ↑ | 49.055 | 50.693 | 41.480 | 50.062 | 42.187 | **51.914** |
| | DD ↓ | 0.650 | 0.527 | 1.517 | 0.561 | 1.395 | **0.448** |
| | ERGAS ↓ | 3.053 | 2.550 | 4.833 | 2.727 | 3.731 | **2.462** |
| | SSIM ↑ | 0.992 | 0.995 | 0.979 | 0.995 | 0.985 | **0.996** |
| | RMSE ↓ | 1.025 | 0.830 | 2.440 | 0.889 | 2.216 | **0.741** |

better contextualize the computational cost of the proposed Algorithm 1, which is given by

$$\begin{aligned}&\mathcal{O}\left(N_{CG}\left(n_w^2 W + n_w W^2\right)\right) + \mathcal{O}\left(N_{CG}\left(n_h^2 H + n_h H^2\right)\right)\\&\quad + \mathcal{O}\left(N_{CG}\left(n_s^2 S + n_s S^2\right)\right)\\&\quad + \mathcal{O}\left(N_{ADMM}\left(n_w^2 n_h n_s + n_w n_h^2 n_s + n_w n_h n_s^2\right)\right)\end{aligned} \tag{21}$$

where $n_w$, $n_h$, and $n_s$ denote the sizes of the core tensor in the Tucker decomposition and $N_{CG}$ and $N_{ADMM}$ denote the number of CG and ADMM iterations, respectively. More details can be found in [12].

Comparing (20) and (21), it is evident that the proposed Algorithm 1 incurs a higher theoretical time complexity per iteration than CSTF. This is primarily because, compared

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TABLE II
HYPERPARAMETER SETTINGS OF THE PROPOSED METHOD ACROSS ALL FOUR TEST DATASETS

| | $R_1$ | $R_2$ | $L_1$ | $L_3$ | $K$ |
|---|---|---|---|---|---|
| Suggested Range | $[2, 11]$ | $[125, 400]$ | $[2, 11]$ | $[2, 4]$ | $[20, 80]$ |
| *CAVE-flowers* | 2 | 400 | 5 | 3 | 50 |
| *Harvard-imgb9* | 6 | 295 | 11 | 3 | 25 |
| *Chikusei* | 10 | 140 | 2 | 2 | 50 |
| *Hyperion-WV3* | 2 | 200 | 2 | 2 | 20 |

with the Tucker format, the TW model explicitly enhances the topological connections among the latent factors to better capture the intrinsic multidimensional correlations.

## IV. NUMERICAL EXPERIMENTS

### A. Datasets

They are given as follows.

1) *CAVE Dataset:*[4] The CAVE dataset has 31 HR-HSIs, each with 31 spectral bands and a spatial resolution of $512 \times 512$ pixels. From this dataset, we randomly chose one HSI, *flowers*, for our experiment. The image was resized to $256 \times 256 \times 31$ to act as the ground truth (GT). The LR-HSI of size $64 \times 64 \times 31$ was then produced by averaging nonoverlapping $4 \times 4$ spatial blocks of the GT. In addition, The HR-MSI of size $256 \times 256 \times 3$ was generated using a Nikon D700 camera.
2) *Harvard Dataset:*[5] The Harvard dataset has 50 HR-HSIs, and each image has 31 spectral bands and a spatial resolution of $1392 \times 1040$ pixels. For our experiments, we randomly chose the image, i.e., *imgb9* and cropped its upper left region to obtain a $512 \times 512 \times 31$ cube. This cube was then scaled down to $256 \times 256 \times 31$ to serve as the GT. The LR-HSI of size $64 \times 64 \times 31$ was produced by averaging nonoverlapping $4 \times 4$ spatial blocks of the GT. Additionally, the three-band HR-MSI was acquired using a Nikon D700 camera.
3) *Chikusei Dataset:*[6] [67] The Chikusei dataset was acquired using the Headwall Hyperspec-VNIR-C sensor, featuring 128 spectral bands ranging from 363 to 1018 nm. The full scene size is $2517 \times 2335$ pixels, with a ground sampling distance (GSD) of 2.5 m. For our experiments, a $256 \times 256 \times 128$ subcube was extracted as the GT. The LR-HSI was obtained by first applying a $7 \times 7$ Gaussian blur (standard deviation 2), followed by downsampling by a factor of 4. The HR-MSI was created by averaging every 32 adjacent spectral bands.

### B. Compared Methods and Quantitative Metrics

*1) Compared Methods:* The proposed CTW method falls within TD-based approaches and, more specifically, within the class of CTD-based methods. Accordingly, we compare CTW with five representative state-of-the-art TD-based techniques, including the UTD-based methods LTTR [56] and GTNN [57], as well as the CTD-based methods STEREO [11], CSTF [12], and CTRF [13]. The code for each method was sourced from the respective authors' websites, and all parameters were carefully tuned following the guidelines provided by the original authors. Experiments were conducted using MATLAB R2025a on a computer equipped with an Intel[7] Core[8] Ultra 7 265KF processor running at 3.90 GHz and 64 GB of RAM.

*2) Quantitative Metrics:* The quality of the fused HR-HSI was evaluated using several quantitative metrics. Our study utilized five indicators: peak signal-to-noise ratio (PSNR), relative dimensionless global error in synthesis (ERGAS) [68], degree of distortion (DD) [69], root-mean-square error (RMSE) [69], and structural similarity index (SSIM) [70]. Higher values of PSNR and SSIM correspond to better fusion quality, whereas lower values of DD, ERGAS, and RMSE indicate improved results.

### C. Experimental Results on Simulated Data Fusion

*1) Quantitative Comparison:* The experimental results on the *CAVE-flowers*, *Harvard-imgb9*, and *Chikusei* datasets are summarized in Table I. As presented in the table, the proposed CTW approach consistently delivers superior fusion performance compared with the competing approaches across the five metrics. On *CAVE-flowers*, CTW outperforms the UTD-based methods LTTR and GTNN by more than 7 and 6 dB in PSNR, respectively. For *Harvard-imgb9*, the PSNR obtained by CTW exceeds that of STEREO and CSTF by more than 1 and 0.7 dB, respectively. On the *Chikusei* dataset, CTW also attains over a 1-dB gain in PSNR compared with the CTD-based method CTRF. Overall, the results reported in Table I illustrate the superiority of the proposed CTW approach.

*2) Visual Comparison:* To provide a more comprehensive comparison among the fusion methods, we further evaluate their performance from a visual perspective. Figs. 6–8 present pseudo-color images reconstructed from selected spectral bands, together with their corresponding error maps, for the *CAVE-flowers*, *Harvard-imgb9*, and *Chikusei* datasets, respectively. Representative regions are marked with red and green boxes to facilitate closer inspection. Overall, the CTD-based methods, i.e., STEREO, CSTF, CTRF, and the proposed CTW, produce visibly lower reconstruction errors than the uncoupled decomposition methods LTTR and GTNN. Among the CTD-based approaches, the proposed CTW yields the smallest errors, highlighting its superiority in HSI-MSI fusion.

### D. Experimental Results on Real Data Fusion

For the real data experiments, LR-HSI was acquired by the Hyperion sensor [71] over the Cuprite mining district in Nevada, USA, on September 19, 2011. The dataset contains 167 bands covering the VNIR and SWIR regions, with a GSD of 30 m, and has spatial dimensions of $927 \times 379$ pixels. The corresponding SWIR HR-MSI was obtained from the WorldView-3 (WV3) satellite over the same area, resampled to a GSD of 7.5 m on September 19, 2014, resulting in

[4]https://www.cs.columbia.edu/CAVE/databases/multispectral/
[5]http://vision.seas.harvard.edu/hyperspec/d2X5g3/
[6]http://naotoyokoya.com/Download.html
[7]Registered trademark.
[8]Trademarked.

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TABLE III
RUNNING TIMES (IN SECONDS) OF ALL THE SIX METHODS ACROSS THE FOUR TEST DATASETS

| Method | STEREO [11] | CSTF [12] | LTTR [56] | CTRF [13] | GTNN[57] | CTW (Ours) |
|---|---|---|---|---|---|---|
| *CAVE-flowers* | 10.615 | 43.311 | 38.945 | 32.997 | 219.979 | 247.233 |
| *Harvard-imgb9* | 9.960 | 67.298 | 36.929 | 33.634 | 243.739 | 1405.000 |
| *Chikusei* | 10.447 | 82.306 | 101.736 | 91.861 | 225.246 | 119.62 |
| *Hyperion-WV3* | 11.937 | 53.222 | 187.567 | 22.830 | 264.858 | 24.092 |

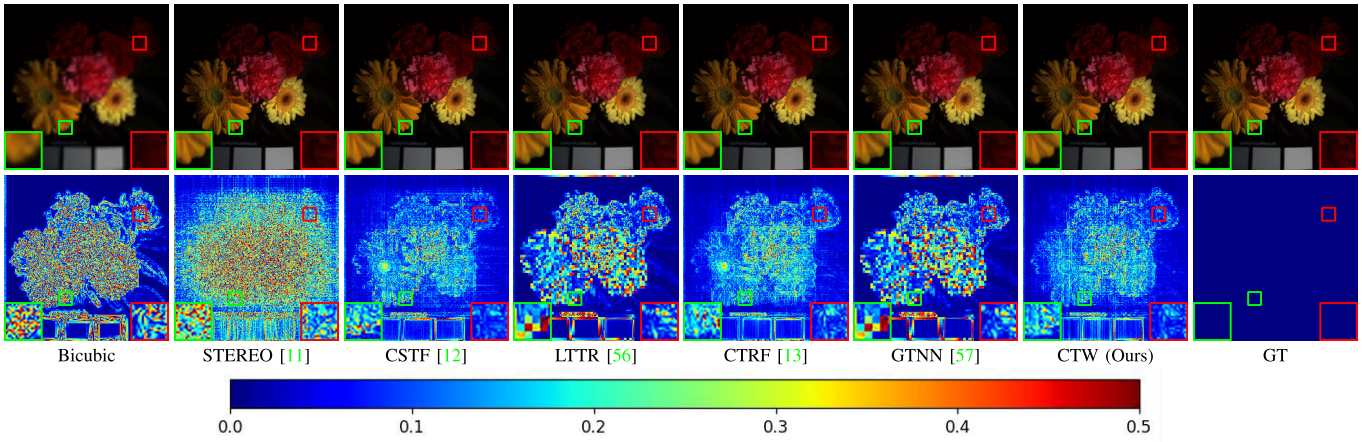



Fig. 6. First row: pseudo-color images consisting of the 30th, 20th, and 10th bands of the fused *flowers*, generated by Bicubic Upsampling, STEREO [11], CSTF [12], LTTR [56], CTRF [13], GTNN [57], CTW, and GT. Second row: the error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 2).

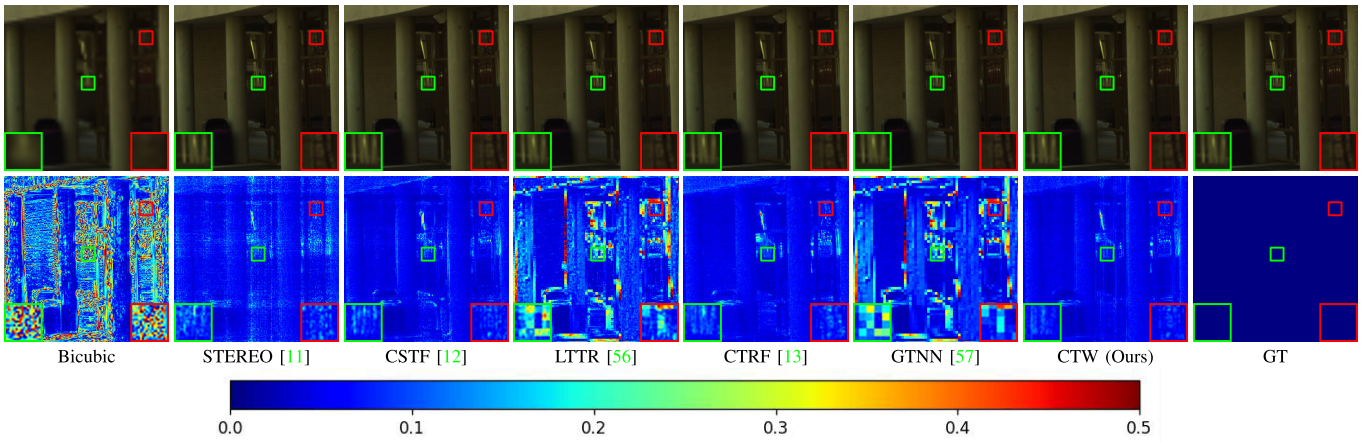



Fig. 7. First row: pseudo-color images consisting of the 30th, 20th, and 10th bands of the fused *imgb9*, generated by Bicubic Upsampling, STEREO [11], CSTF [12], LTTR [56], CTRF [13], GTNN [57], CTW, and GT. Second row: the error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 2).

an image of size $3710 \times 1518 \times 8$. From these datasets, a $128 \times 128 \times 167$ subimage was extracted from the Hyperion data and the corresponding coregistered $512 \times 512 \times 8$ subimage was taken from the WV3 data. These were then resized to $64 \times 64 \times 167$ and $256 \times 256 \times 8$, respectively, to serve as LR-HSI and HR-MSI inputs. The spatial and spectral responses were estimated following the procedure in [41]. Fig. 9 displays pseudo-color images produced by all competing fusion methods, with key regions marked by red and green boxes for closer comparison. As shown, the proposed CTW method, along with CTRF, CSTF, and STEREO, preserves fine spatial structures and generates visually superior results relative to LTTR and GTNN. These results validate the practical effectiveness of the CTW method in real data fusion tasks.

## V. DISCUSSION

### A. Hyperparameter Analysis

Here, we analyze the effects of hyperparameters in Algorithm 1. The parameters examined include the TW-ranks $\mathbf{r} = [R_1, R_2, R_3, L_1, L_2, L_3]$, the positive hyperparameter $\rho$, and the number of iterations $K$. Throughout our experimental studies, we empirically set $R_3 = R_1$ and $L_2 = L_1$ and fix $\rho = 1$ for all test datasets. The discussion focuses on results obtained using the *Chikusei* dataset (see Section IV-A).

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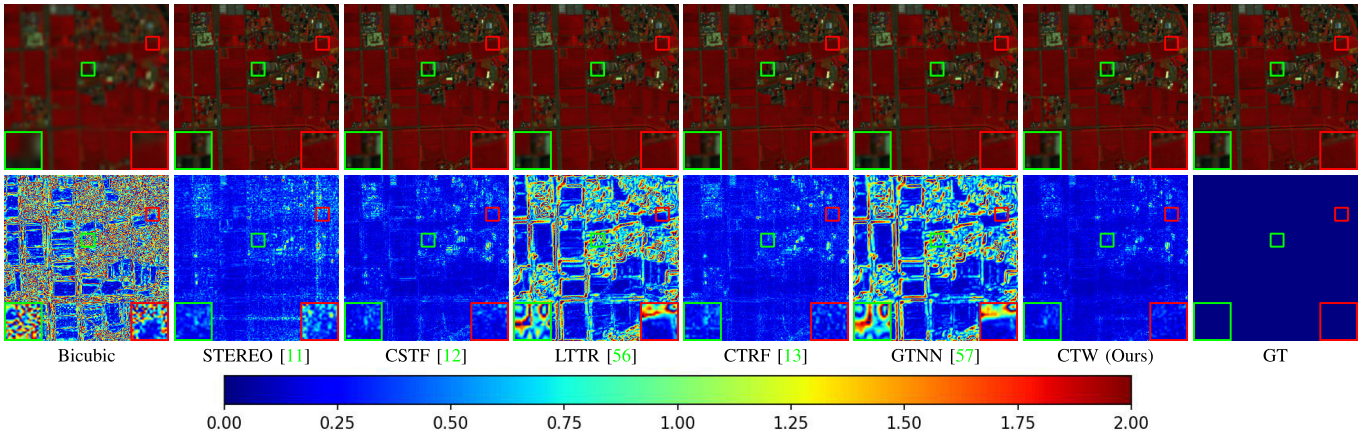



Fig. 8. First row: pseudo-color images consisting of the 120th, 60th, and 10th bands of the fused *Chikusei*, generated by Bicubic Upsampling, STEREO [11], CSTF [12], LTTR [56], CTRF [13], GTNN [57], CTW, and GT. Second row: the error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 0.5).

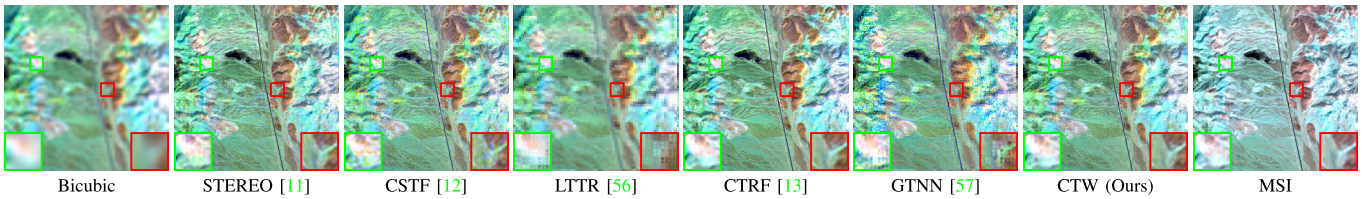



Fig. 9. Pseudo-color images consisting of the 2.20-$\mu$m red(152nd), 1.52-$\mu$m (103rd), and 0.57-$\mu$m (15th) bands of fused Hyperion and WV3 data, generated by Bicubic, STEREO [11], CSTF [12], LTTR [56], CTRF [13], GTNN [57], and CTW. WV3 (MSI) (the 2.32 $\mu$m (8th), 1.57 $\mu$m (2nd), and 1.21 $\mu$m (1st) bands).

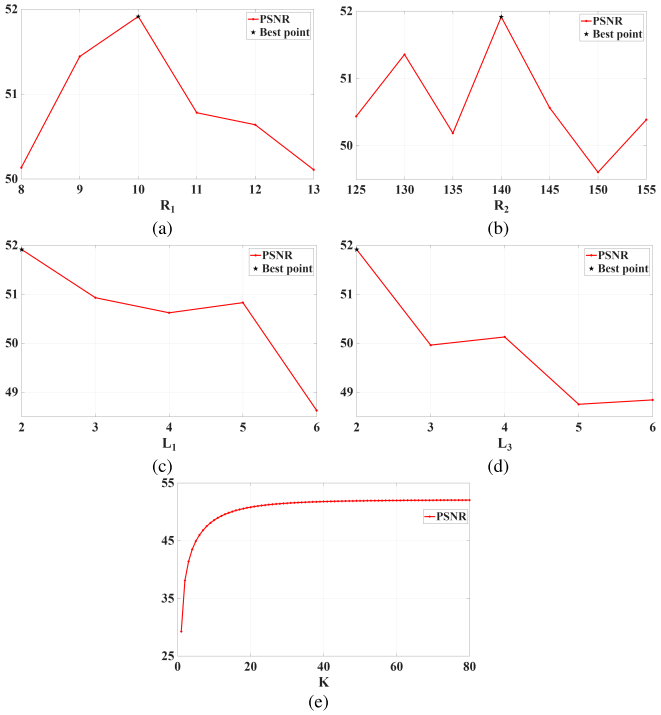



Fig. 10. Parameter analysis of Algorithm 1. (a) $R_1$ ($R_3 = R_1$), (b) $R_2$, (c) $L_1$ ($L_2 = L_1$), (d) $L_3$, and (e) $K$. (Dataset: *Chikusei*.)

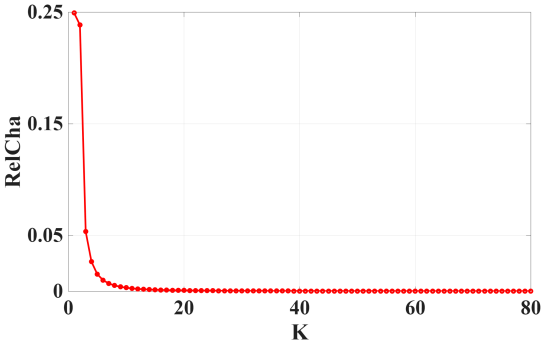



Fig. 11. Numerical convergence curve. (Dataset: *Chikusei*.)

*1) Analysis of the TW-Ranks* $\mathbf{r}$*:* The tensor rank is closely tied to the intrinsic structure of the tensor, making the determination of an optimal rank a complex problem within the TD community. Typically treated as a hyperparameter in TD approaches, selecting the best tensor rank often requires balancing multiple factors, including but not limited to accuracy. In this article, we utilized a PSNR-based approach to identify the optimal TW-ranks, focusing on parameters $R_1$, $R_2$, $L_1$, and $L_3$. Candidate ranges were set as $[8, 13]$ for $R_1$, $[125, 155]$ for $R_2$, and $[2, 6]$ for both $L_1$ and $L_3$. Fig. 10(a) shows how PSNR varies with $R_1$ between 8 and 13, with a sharp increase up to a peak at $R_1 = 10$, followed by a gradual decline. Accordingly, $R_1 = 10$ was selected. In Fig. 10(b), PSNR fluctuations indicate that Algorithm 1 is sensitive to $R_2$, which attains its maximum value at 140, leading to the choice of $R_2 = 140$. Fig. 10(c) and (d) depicts PSNR trends for $L_1$ and $L_3$ within the range of 2–6, both showing a decline as

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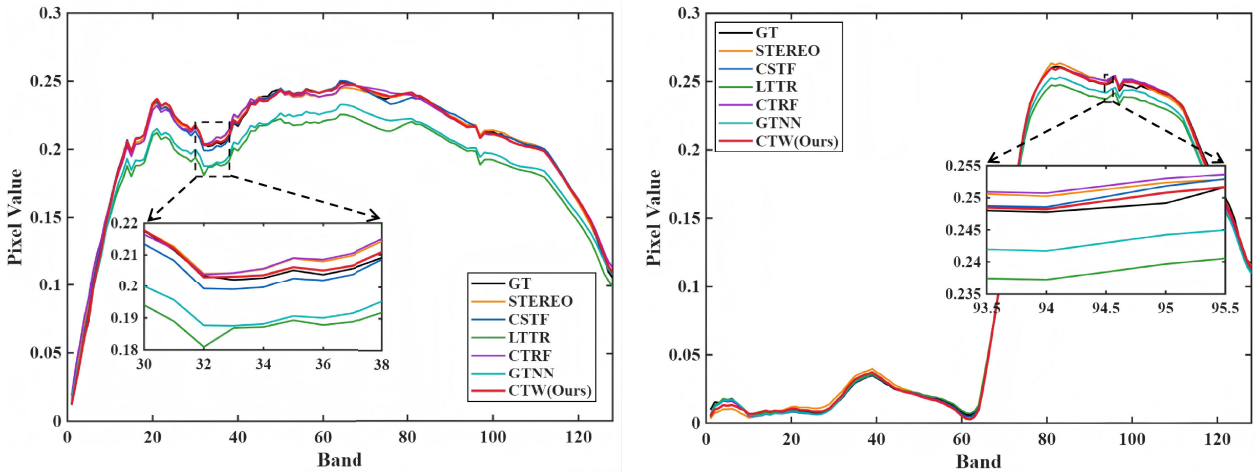



Fig. 12. Spectral curves between fused results and GT at different pixels. Left: (20, 20). Right: (100, 140). (Dataset: *Chikusei*.)

the parameters increase. Based on this observation, $R_3$ and $R_4$ were both set to 2.

*2) Analysis of the Number of Iterations K:* Fig. 10(e) presents how PSNR changes as the $K$ increases. The plot reveals a rapid rise in PSNR from $K = 1$ to $K = 20$, followed by a gradual increase between $K = 20$ and $K = 60$, after which it stabilizes. Considering both fusion quality and running time, we chose to set $K$ at 50.

For the *CAVE-flowers* and *Harvard-imgb9* datasets, we applied the same approach used for the *Chikusei* dataset to identify the optimal hyperparameters. For the real data experiments, the best hyperparameter values were chosen based on visual assessment. The suggested ranges for hyperparameters and the final values selected for all four datasets are summarized in Table II.

### B. Running Time

Table III reports the running times of all the six fusion algorithms on the four test datasets. As shown in the table, the coupled CP-based method STEREO is the fastest, which is consistent with the simplicity of the CP model and its relatively low per-iteration cost. GTNN requires substantially more computation time because its optimization procedure relies on repeated SVDs. For CSTF, LTTR, CTRF, and CTW, the running time depends on the selected tensor ranks and the number of iterations: larger ranks or more iterations naturally increase the computational cost. The particularly long running time of CTW on the *Harvard-imgb9* dataset is mainly due to the update of fourth-order spatial factors and spectral factors in the TW format, combined with the relatively large TW-ranks required for accurate representation on this dataset. On the remaining datasets, however, the running time of CTW is comparable to that of the other methods. Overall, CTW delivers superior fusion performance (as discussed in Sections IV-C and IV-D) while maintaining competitive running times.

### C. Numerical Convergence Behavior

We conducted experiments to evaluate the numerical convergence behavior of the proposed Algorithm 1. Fig. 11 presents the changes in RelCha (RelCha $= (\|\mathcal{X}^{(k+1)} - \mathcal{X}^{(k)}\|_F)/(\|\mathcal{X}^{(k)}\|_F)$, where $\mathcal{X}^{(k+1)}$ and $\mathcal{X}^{(k)}$ represent the outputs from the $(k+1)$th and the $k$th iteration in Algorithm 1, respectively) as the number of iterations $K$ increases on the *Chikusei* dataset. The results show that RelCha decreases steadily and eventually stabilizes, demonstrating the numerical convergence of the proposed Algorithm 1.

### D. Comparison of Spectral Signatures

To visually demonstrate the superiority of the proposed algorithm in preserving spectral fidelity, this section presents the spectral signatures of the fusion results generated by six comparative methods alongside the GT at two distinct pixel locations, as depicted in Fig. 12. For a clearer comparison, representative regions within the curves have been locally magnified. It is evident from the figure that the spectral profiles produced by CTRF, CSTF, STEREO, and the proposed CTW method closely align with the GT at both locations. In contrast, the results from LTTR and GTNN exhibit significant deviations. Furthermore, the spectral curves generated by our proposed CTW method achieve the highest degree of consistency with the GT, validating its advantage in maintaining spectral fidelity.

## VI. Conclusion

In this work, we proposed the CTW method for the HSI-MSI fusion. Building on the strong capability of the TW decomposition to capture intrinsic tensor correlations and extract features, we incorporated the TW format into the HSI-MSI fusion framework and constructed a CTW model. A PAM-based algorithm was designed to solve the CTW model, along with an analysis of its computational complexity and numerical convergence. Evaluations on simulated and real datasets indicated that CTW provides improved fusion performance compared to the latest methods. Moreover, despite the more accurate representation provided by the TW format, the CTW maintains competitive running times across most test datasets. These findings validate both the effectiveness and efficiency of the proposed CTW approach.

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Nevertheless, the optimization process on some datasets with complex spatial–spectral correlations, such as *Harvard-imgb9*, still incurs relatively high computational cost. Future work will focus on developing computationally efficient tensor representations and accelerated optimization strategies to achieve a better tradeoff between fusion performance and computational complexity.

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